JYOTHIR RAGHAVALU BHOGI

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Jyothir-dev (Jyothir) (github.com)

EDUCATION

Lovely Professional University — Jalandhar, India
Bachelor of Technology (Computer Science and Engineering) — **July 2022 - May 2026**

SKILLS SUMMARY

- **Programming**: Python, SQL, MATLAB (familiarity)
- **Machine Learning**: Machine Learning, Deep Learning, PyTorch, Large Language Models (LLMs), Generative AI
- **Data**: Data Handling, Data Processing, Dataset Preprocessing, Data Validation, ETL Pipelines, JSON
- **Automation**: Python Automation Scripting, Workflow Automation, AI-Driven Automation
- **Engineering**: Engineering Analysis Support, Experimentation Workflows, Scientific Computing
- **Version Control**: Git, GitHub
- **Development**: Clean Code Development, Debugging, REST APIs, Software Development Life Cycle (SDLC)
- **Systems**: Linux, Red Hat Enterprise Linux (RHEL), Docker
- **Documentation**: LaTeX, Technical Documentation
- **Collaboration**: Analytical Problem Solving, Cross-Functional Collaboration, R&D Environment

WORK EXPERIENCE

DATA ENGINEER INTERN | flickDONE | LINK — **June 25- August 25**

- Optimized Python ETL pipelines processing 50,000+ records daily, improving automation reliability and validation workflows.
- Engineered SQL schemas and indexing strategies reducing query latency by 35% across distributed environments.
- Developed Python automation scripts and REST API validation tools enabling reliable data handling workflows.
- Collaborated with cross-functional R&D teams using Git delivering scalable, maintainable engineering software solutions.
- Implemented Docker-based CI/CD pipelines enabling automated testing environments improving deployment stability.

PROJECTS

Generative AI Product Recommender System | LINK — **June 25- August 2025**

- Developed AI-powered recommendation engine using Python and SQL enabling intelligent product discovery.
- Integrated Large Language Models applying machine learning techniques improving contextual recommendation accuracy.
- Implemented Docker containers and CI/CD pipelines improving testing environments and deployment reliability.
- Designed modular REST API validation scripts supporting structured workflows and reproducible validation experiments.

Rails TODO App | LINK — **September 24- November 2024**

- Developed scalable task management platform using Ruby on Rails MVC architecture.
- Implemented CRUD operations, authentication systems, and database validation ensuring maintainable backend architecture.
- Optimized SQL queries improving response time and application scalability supporting more than 1,000 users.
- Ensured maintainable architecture following Agile practices supporting debugging, testing, and efficient feature delivery.

Morning Market Brief | LINK — **June 24- August 2024**

- Developed cloud-based analytics platform using Python pipelines supporting automated engineering analysis insights.
- Built ETL workflows with Python, SQL, JSON enabling scalable data processing and dataset validation.
- Implemented REST API automation testing frameworks ensuring accurate data handling and validation across workflows.
- Integrated Docker containers and CI/CD pipelines improving system availability and deployment reliability.

RESEARCH AND PUBLICATIONS

Multi-Stage Ensemble for Driver Fatigue Recognition | IEEE, 2025

- Developed PyTorch ensemble deep learning pipeline enabling automated fatigue detection achieving 92% model accuracy.
- Optimized feature extraction and preprocessing reducing computational overhead improving machine learning performance.
- Applied structured experimentation and hyperparameter tuning enabling reproducible machine learning validation pipelines.

CERTIFICATES

Python, Data Science & ML (Cipherschools) | CERTIFICATE — **August 2024**

- Mastered Python programming including automation scripting, object-oriented design, and scalable development practices.
- Applied machine learning algorithms, feature engineering techniques, and analytical problem solving on real-world datasets.

Generative AI with LLMs (Coursera) | CERTIFICATE — **March 2024**

- Learned transformer architectures, prompt engineering, and core concepts behind modern Large Language Model systems.
- Built practical generative AI applications enabling document summarization, search automation, and intelligent workflows.